

Business Intelligence Lab

Unit 1 | Session 2

Experiment 2

Designing and Implementing ETL Processes using Microsoft Power BI

Aim

To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor (Built into Power BI Desktop)
4. DAX (Data Analysis Expressions)

Dataset Source:

Global Superstore Dataset (Kaggle)

<https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Theory

ETL (Extract, Transform, Load) is one of the most important processes in Business Intelligence. It enables organizations to collect data from source systems, transform it into a clean and consistent format, and load it into a centralized environment for analysis.

Extract Phase:

The extraction phase involves collecting data from source systems. In this experiment, data is extracted from the Orders, Returns, and People worksheets available in the Global Superstore dataset.

Transform Phase:

The transformation phase includes data cleaning, handling missing values, correcting data types, creating derived columns, filtering invalid records, and integrating multiple datasets.

Load Phase:

The transformed data is loaded into the Power BI data model where relationships can be established and analytical measures can be created using DAX.

Power Query Editor serves as the ETL engine in Power BI and records each transformation step as an Applied Step, making the workflow transparent and reproducible.

Conduct

Part A: Extract – Connect to Dataset

Step 1: Download the Global Superstore Dataset from Kaggle.

Step 2: Open Power BI Desktop.

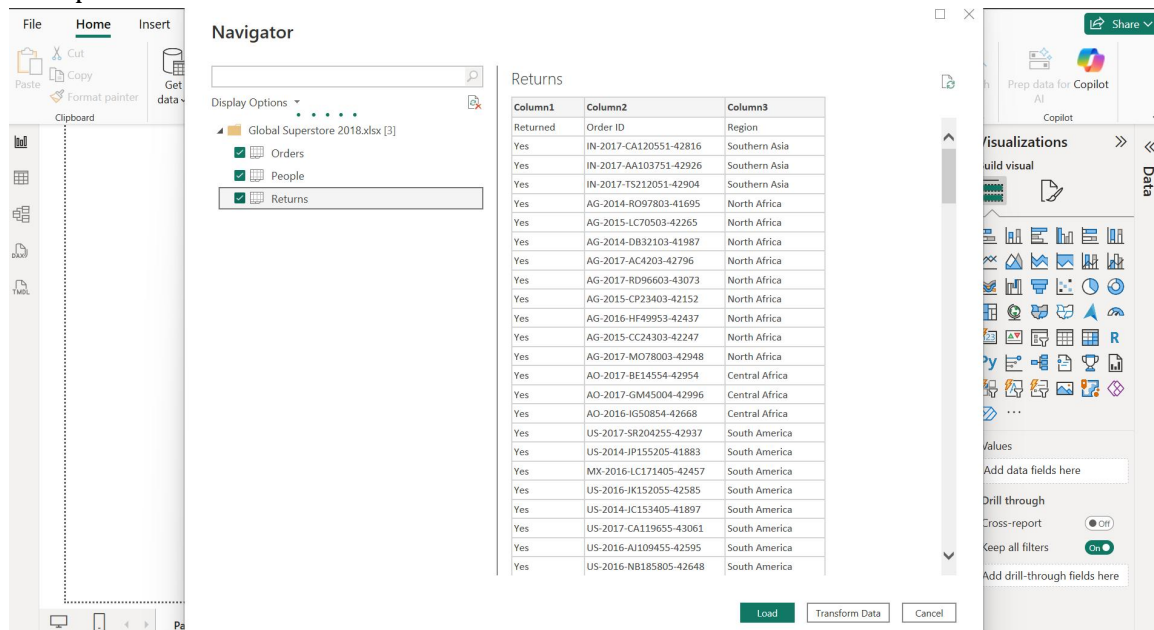
Home > Get Data > Excel Workbook

Select Global Superstore Dataset.xlsx

Click Transform Data.

Step 3: Import the following worksheets:

- Orders
- Returns
- People

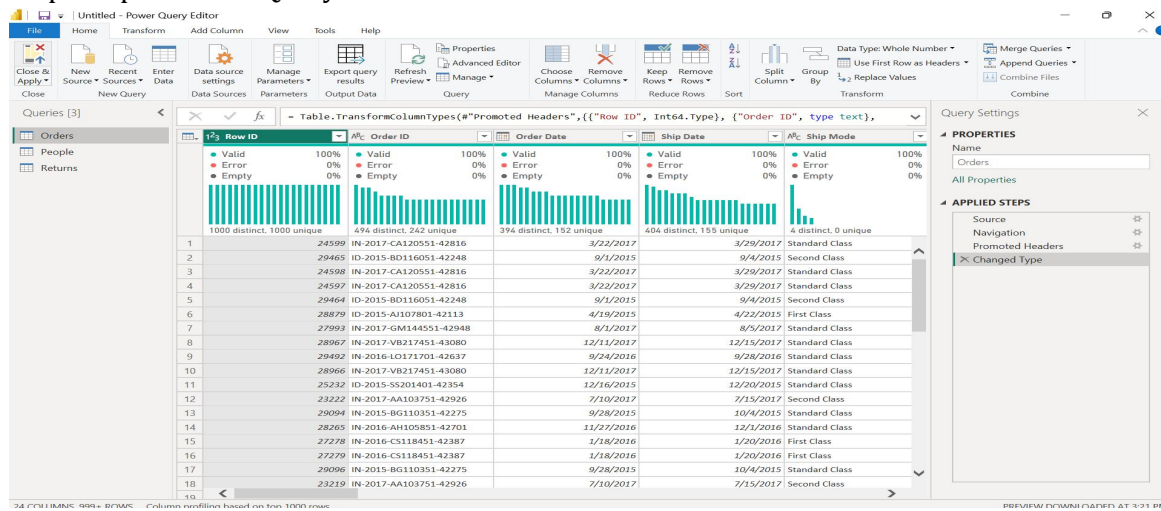


Column1	Column2	Column3
Returned	Order ID	Region
Yes	IN-2017-CA120551-42816	Southern Asia
Yes	IN-2017-AA103751-42926	Southern Asia
Yes	IN-2017-TS212051-42904	Southern Asia
Yes	AG-2014-RO97803-41695	North Africa
Yes	AG-2015-LC70503-42265	North Africa
Yes	AG-2014-DB32103-41987	North Africa
Yes	AG-2017-AC4203-42796	North Africa
Yes	AG-2017-RD96603-43073	North Africa
Yes	AG-2015-CP23403-42152	North Africa
Yes	AG-2016-HF49953-42437	North Africa
Yes	AG-2015-CC24303-42247	North Africa
Yes	AG-2017-MO78003-42948	North Africa
Yes	AO-2017-BE14554-42954	Central Africa
Yes	AO-2017-GM45004-42996	Central Africa
Yes	AO-2016-IG50854-42668	Central Africa
Yes	US-2017-SR204255-42937	South America
Yes	US-2014-JP155205-41883	South America
Yes	MX-2016-LC171405-42457	South America
Yes	US-2016-JK152055-42585	South America
Yes	US-2014-JC153405-41897	South America
Yes	US-2017-CA119655-43061	South America
Yes	US-2016-AJ109455-42595	South America
Yes	US-2016-NB185805-42648	South America

Verify that all imported tables appear in the Queries pane.

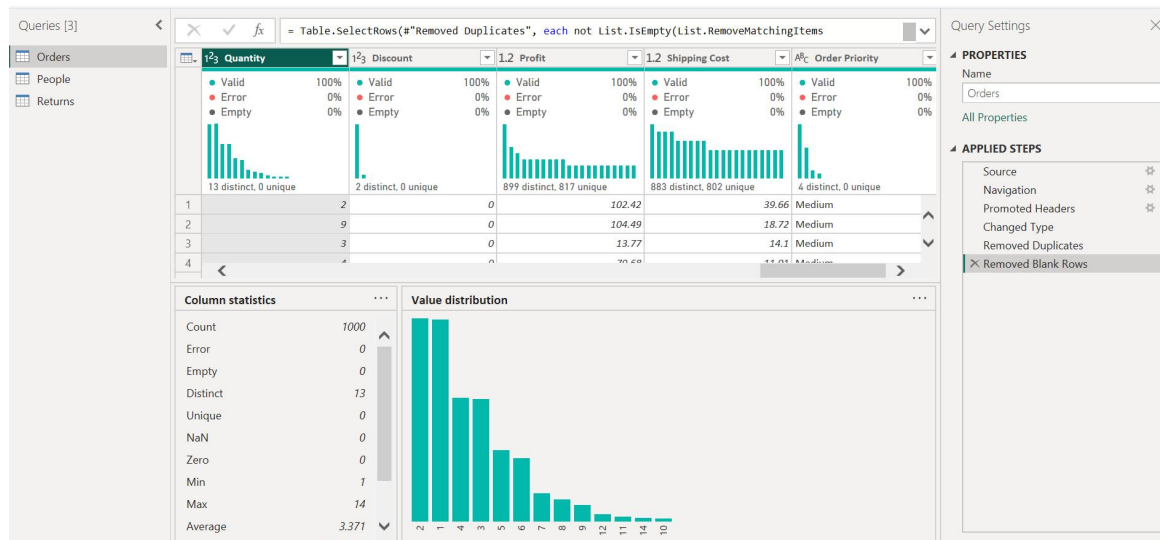
Part B: Transform – Power Query Editor

Step 4: Open Power Query Editor.

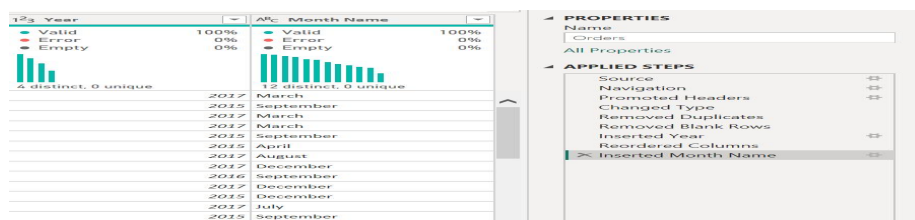


Step 5: Apply the following transformations to the Orders table:

- Remove duplicate records
- Remove blank rows
- Verify and correct data types
- Rename columns where required



Extract Date and Year



- Create a custom Profit Margin column
- Standardize Region and Segment values

Step 6: Transform the Returns table:

- Remove duplicate Order IDs
- Check for missing values
- Standardize Returned field values

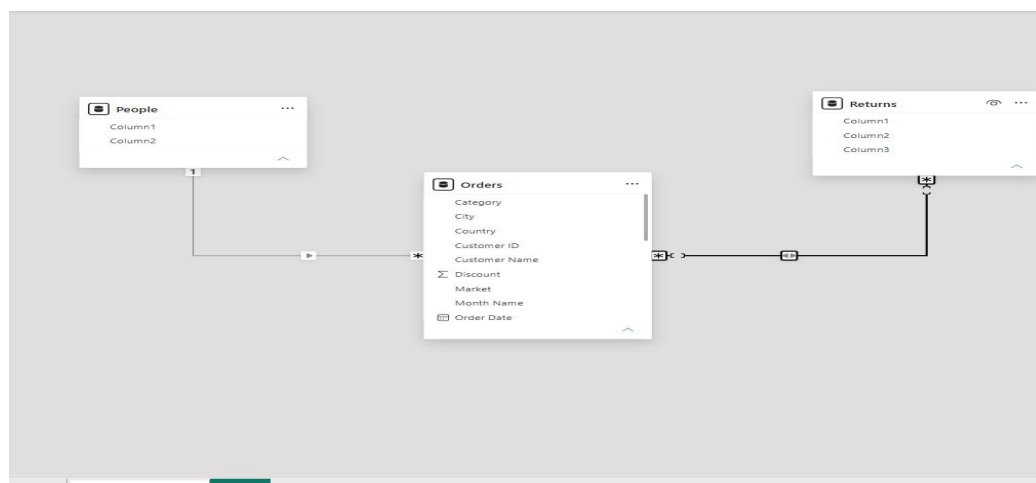
Step 7: Transform the People table:

- Validate regional assignments
- Remove unnecessary columns
- Verify consistency of region names

Node	Customer ID	Customer Name	Segment
1	3/29/2017	Standard Class	CA-120551
2	9/4/2015	Second Class	BD-116051
3	3/29/2017	Standard Class	CA-120551
4	3/29/2017	Standard Class	CA-120551
5	9/4/2015	Second Class	BD-116051
6	4/22/2015	First Class	AJ-107801
7	8/5/2017	Standard Class	GM-144551
8	12/15/2017	Standard Class	VB-217451
9	9/28/2016	Standard Class	LO-171701
10	12/15/2017	Standard Class	VB-217451
11	12/20/2015	Standard Class	SS-201401
12	7/15/2017	Second Class	AA-103751
13	10/4/2015	Standard Class	BG-110351
14	12/1/2016	Standard Class	AH-105851

Step 8: Merge Orders and Returns tables using Order ID.

Step 9: Review all Applied Steps and click Close & Apply.



Part C: Load – Build Data Model

Step 10: Navigate to Model View.

Step 11: Create relationships:

Orders ↔ Returns (Order ID)

Orders ↔ People (Region)

Step 12: Create DAX measures:

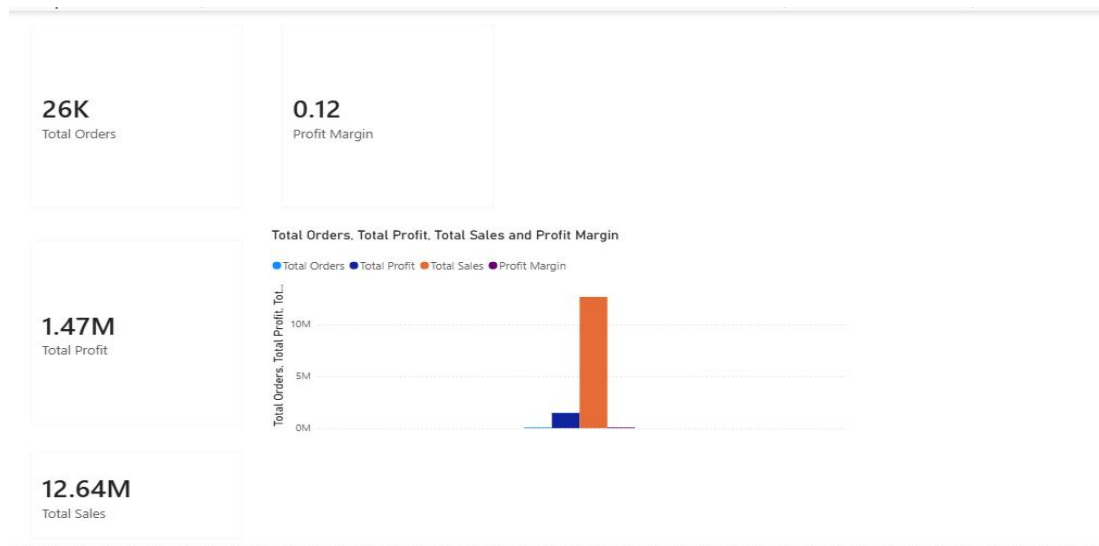
Total Sales = SUM(Orders[Sales])

Total Profit = SUM(Orders[Profit])

Profit Margin = DIVIDE(SUM(Orders[Profit]),SUM(Orders[Sales]))

Total Orders = DISTINCTCOUNT(Orders[Order ID])

Step 13: Create a simple report visual using fields from related tables.



Result

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

Learning Outcomes

After completing this experiment, students will be able to:

- Explain the ETL process and its role in Business Intelligence systems. (CO1-BT2)
- Extract data from real-world business datasets using Power BI. (CO2-BT3)
- Apply transformation techniques including cleaning, filtering, and merging. (CO2-BT3)
- Create relationships and analytical measures in Power BI. (CO3-BT4)
- Validate transformed datasets for reporting and decision-making. (CO3-BT4)